

Physical Examples of First-Order Systems

Liquid Level

Tank of uniform cross sectional area A

flow resistance R — valve

$q_0 \rightarrow$ volumetric flow rate

$h \rightarrow$ head

$$\left[\frac{\text{m}^3}{\text{sec}} \right] q_0 = \frac{h \text{ [m]}}{R \left[\frac{\text{sec}}{\text{m}^2} \right]}$$

$$R = \frac{h}{q_0} \Rightarrow \text{Linear resistance (e.g pipe)}$$

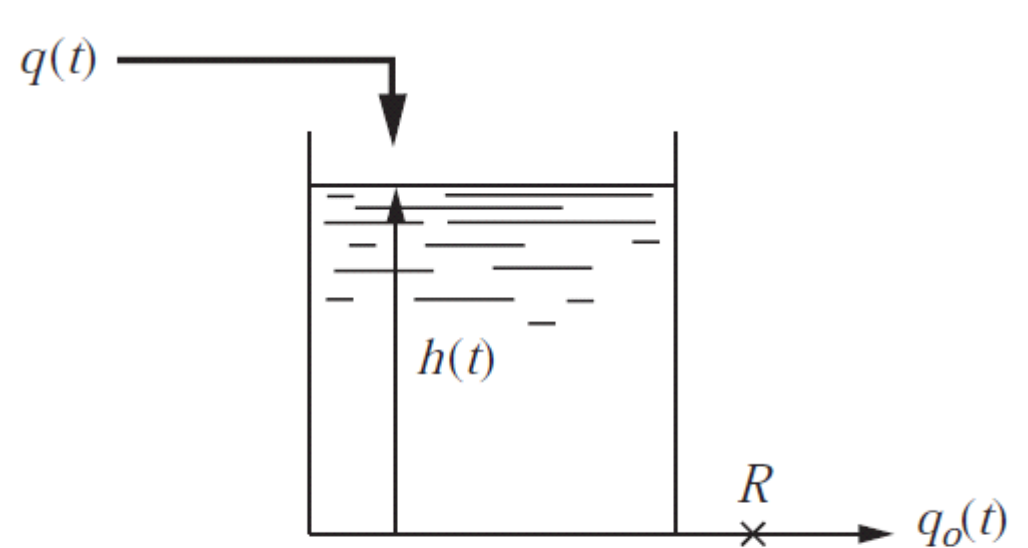


FIGURE 5-1
Liquid-level system.

Flow generally proportional to $\sqrt{h}$ in pipes

Time varying volumetric flow q of liq

Transient mass balance

Mass flow in - Mass flow out = Accumulation in tank

$$\int q(t) - \int q_o(t) = \frac{d}{dt} (SAh)$$

$\frac{h''}{R}$

$$q - \frac{h}{R} = A \frac{dh}{dt}$$

For Steady State $\frac{dh}{dt} \rightarrow 0$

$$q_s - \frac{h_s}{R} = 0$$

Difference from steady state

$$(q - q_s) = \frac{1}{R} (h - h_s) + A \frac{d(h - h_s)}{dt}$$

Using deviation variables for flow rate

$$Q = q - q_s$$

Height

$$H = h - h_s$$

Gives

$$Q = \frac{1}{R} H + A \frac{dH}{dt}$$

Taking Laplace transform

$$Q(s) = \frac{1}{R} H(s) + A s H(s) + H(0)$$

Forming first order lag

$$Q(s) = H(s) \left(\frac{1}{R} + A s \right) = H(s) \frac{1 + ARs}{R}$$

$$\frac{H(s)}{Q(s)} = \frac{R}{1 + ARs}$$

$$; AR = \tau$$
$$[m^2] \left[\frac{sec}{m^2} \right] = [sec]$$

Changing flow rate with unit step function

$$Q(t) = u(t)$$

$$Q(s) = \frac{1}{s}$$

$$H(s) = \frac{1}{s} \frac{R}{\tau s + 1}$$

$$H(t) \Big|_{t \rightarrow \infty} = \lim_{s \rightarrow 0} [s H(s)] = \lim_{s \rightarrow 0} \frac{R}{\tau s + 1} = R$$

Ultimate change is R

How to relate inlet and outlet flow

$$\text{as } q_0 = \frac{h}{R} \Rightarrow q_{0s} = \frac{h_s}{R}$$

$$q_0 - q_s = \frac{h - h_s}{R}$$

$$Q_0 = \frac{H}{R}$$

Transforms

$$Q_0(s) = \frac{H(s)}{R}$$

Combining with

$$\frac{H(s)}{Q(s)} = \frac{R}{\tau s + 1}$$

Plug $H(s) = R Q_0(s)$

$$\boxed{\frac{Q_0(s)}{Q(s)} = \frac{1}{\tau s + 1}}$$

Implications of impulse input

Example

$$\text{Time const} = 1 \text{ min}$$

$$\text{Resistance} = \frac{1}{9} \frac{\text{ft}}{\text{cfm}}$$

$$Q_0 = 10 \text{ ft}^3/\text{min} \text{ (cfm)}$$

@ $t=0$ flow increases to $100 \text{ ft}^3/\text{min}$
for 0.1 min

$$\text{Volume added} = 9 \text{ ft}^3$$

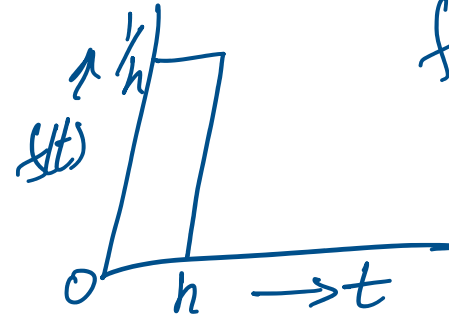
Input approaches impulse func

$$\frac{H(s)}{Q(s)} = \frac{R}{s+1} = \frac{1/9}{s+1}$$

Input is difference of step functions

$$Q(t) = (100 - 10) [u(t) - u(t - 0.1)]$$

Impulse func



$$f(t) = \frac{1}{h} [u(t) - u(t-h)]$$

$$f(s) = \frac{1}{h} \frac{1 - e^{-hs}}{s}$$

$$Q(s) = \frac{90}{s} (1 - e^{-0.1s})$$

$$H(s) = \frac{1/9}{s+1} \left[\frac{90}{s} (1 - e^{-0.1s}) \right]$$

$$= 10 \left[\frac{1}{s(s+1)} - \frac{e^{-0.1s}}{s(s+1)} \right]$$

$$H(s) = 10 \left[\frac{1}{s^2 + s} - \frac{e^{-0.1s}}{s(s+1)} \right]$$

$\downarrow$ $\downarrow$
 $1 - e^{-t}$ Inverse of $e^{-st_0} f(s) = f(t-t_0)u(t-t_0)$

with $u(t-t_0) = 0$ for $t-t_0 < 0$
 $\Rightarrow t < t_0$

$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{e^{-0.1s}}{s(s+1)} \right\} = 0 \quad \text{for } t < 0.1$$

$$= 10(1 - e^{-(t-0.1)}) \quad \text{for } t > 0.1$$

$$\Rightarrow 10(1 - e^{-(t-0.1)}) u(t-0.1)$$

$$H(t) = 10(1 - e^{-t}) u(t) - 10(1 - e^{-(t-0.1)}) u(t-0.1)$$

$$H(t) = 10(1 - e^{-t}) \quad t < 0.1$$

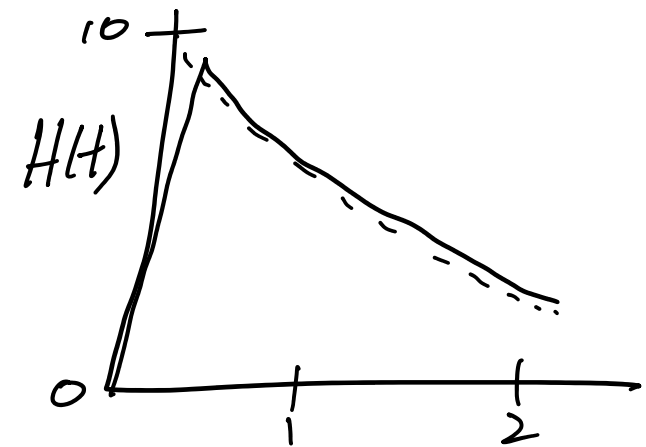
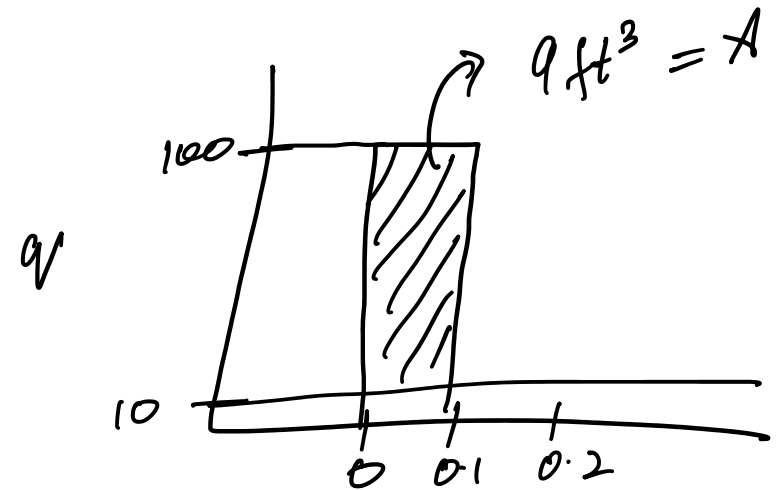
$$H(t) = 10[(1 - e^{-t}) - (1 - e^{-(t-0.1)})] \quad t > 0.1$$

$$\begin{aligned}
 H(t) &= 10 [(1 - e^{-t}) - (1 - e^{-(t-0.1)})] \\
 &= 10 [1 - e^{-t} - 1 + e^{-(t-0.1)}] \\
 &= 10 [-e^{-t} + e^{-(t-0.1)}] \\
 &= 10 [-e^{-t} + e^{-t} e^{0.1}] \\
 &= 10 [e^{-t} (-1 + e^{0.1})] \\
 &= 10 [e^{-t} 0.105]
 \end{aligned}$$

$$H(t) = 1.052 e^{-t} \quad \text{for } t > 0.1$$

Impulse magnitude is q

$$H(t) |_{\text{impulse}} = \left(\frac{1}{q} e^{-t} \right) \times \underset{\substack{\uparrow \\ Q(1s) = q - q/2}}{q} = e^{-t}$$



Liquid level Process with constant - flow outlet

- using constant flow pump
- const cross sectional Area
- const density

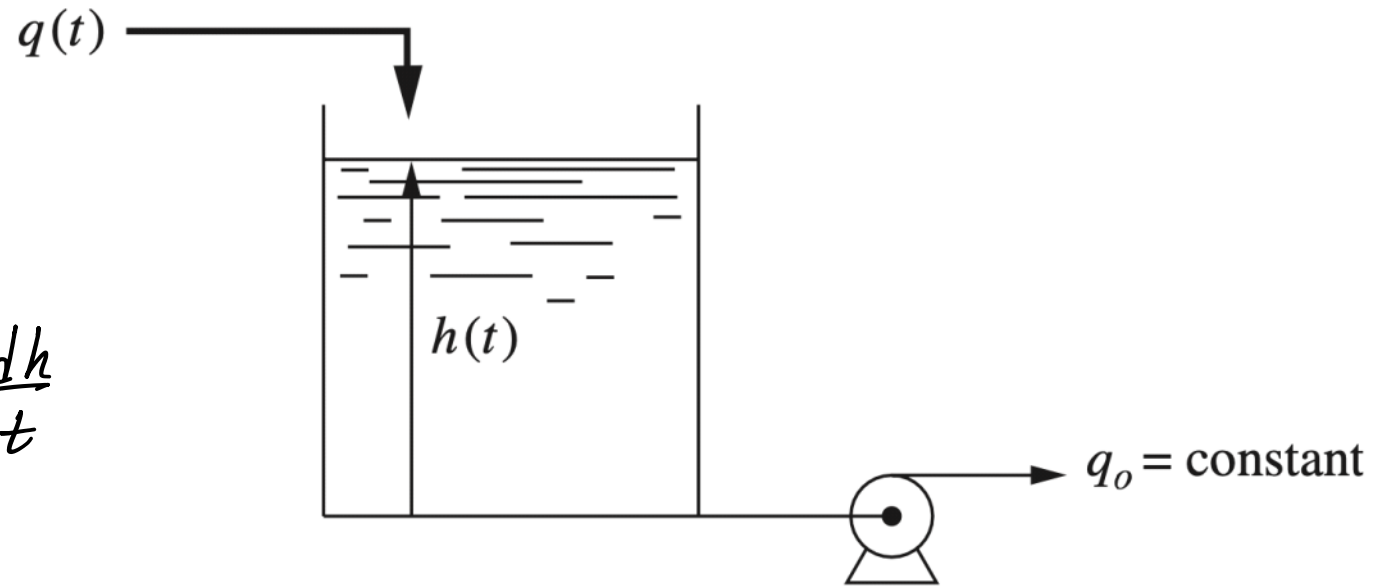
Unsteady Mass Balance

$$q_i(t) - q_o(t) = A \frac{dh}{dt}$$

$\uparrow$
now const

$$q_i(t) - q_o = A \frac{dh}{dt}$$

$$@ \text{ S.S } \Rightarrow q_s - q_o = 0$$



Using deviation variables

$$Q = A \frac{dH}{dt}$$

Laplace transform

$$Q(s) = A(sH(s) - \overset{\circ}{H(0)})$$

$$\frac{H(s)}{Q(s)} = \frac{1}{As}$$

$$H(s) = \frac{1}{As} Q(s)$$

$$h(t) - h_s = \frac{1}{A} \int_0^t Q(t) dt$$

$$h(t) = h_s + \frac{1}{A} \int_0^t Q(t) dt$$

if $Q(t)$ is $u(t) \Rightarrow h(t) = h_s + \frac{t}{A}$
Step change

This function will grow without limit and is termed "non-regulation"

Where there is limited change in output for sustained change in input; system is "regulated"

for R @ outlet T.F

$$\frac{R}{Ts+1} = \frac{R}{ARs+1}$$

$$\lim_{R \rightarrow \infty} \left(\frac{R}{ARs+1} \right) = \frac{1}{As + 1/R}$$
$$= \frac{1}{As}$$

Mining Process as first order system

Input containing dissolved salts enters @ constant volumetric flow rate

Salt conc. varies with time

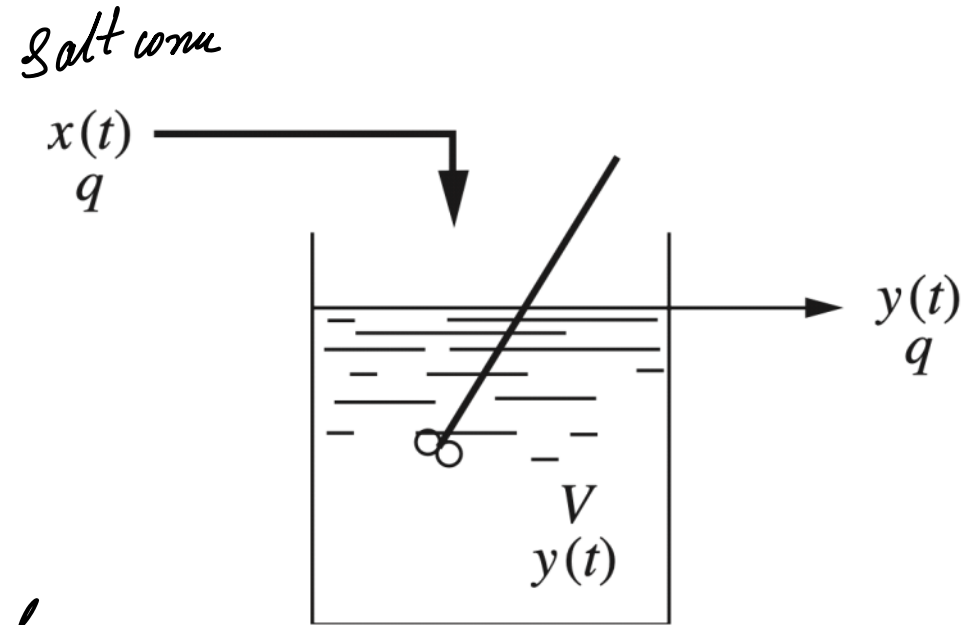
Relation b/w outlet and inlet conc.?

⇒ Consider no change in density

Salt input flow rate - Salt output flow rate = Accumulation of salt in tank

$$q \cdot x - q \cdot y = \frac{d(Vy)}{dt} = V \frac{dy}{dt}$$

$$@ \text{ S.S } \Rightarrow q x_s - q y_s = 0$$



Using deviation variables

$$qX - qY = V \frac{dY}{dt}$$

$$q(X(s) - Y(s)) = V s Y(s) - \cancel{Y(0)}$$

$$qX(s) = qY(s) + V s Y(s)$$

$$\frac{Y(s)}{X(s)} = \frac{q}{q + Vs}$$

$$= \frac{1}{\frac{V}{q}s + 1}$$

$$\frac{Y(s)}{X(s)} = \frac{1}{\tau s + 1}$$

$$\tau = \frac{V}{q} = \frac{\frac{[m^3]}{[m^3]}}{\frac{[m^3]}{[sec]}} = [sec]$$

Heating Process as first order

Const flow rate w

$T_i \rightarrow$ input ; $T \rightarrow$ output

Rate of heat in - Rate of heat out + Rate of in from heater = Rate of accumulation of heat

$$wC(T_i - T_{ref}) - wC(T - T_{ref}) + q = \rho VC \frac{d(T - T_{ref})}{dt}$$

$\downarrow$
heat capacity $\Rightarrow$ const

$$= \rho VC \frac{dT}{dt}$$

@ S.S

$$wC(T_{is} - T_s) + q_s = 0$$

Subtracting the two

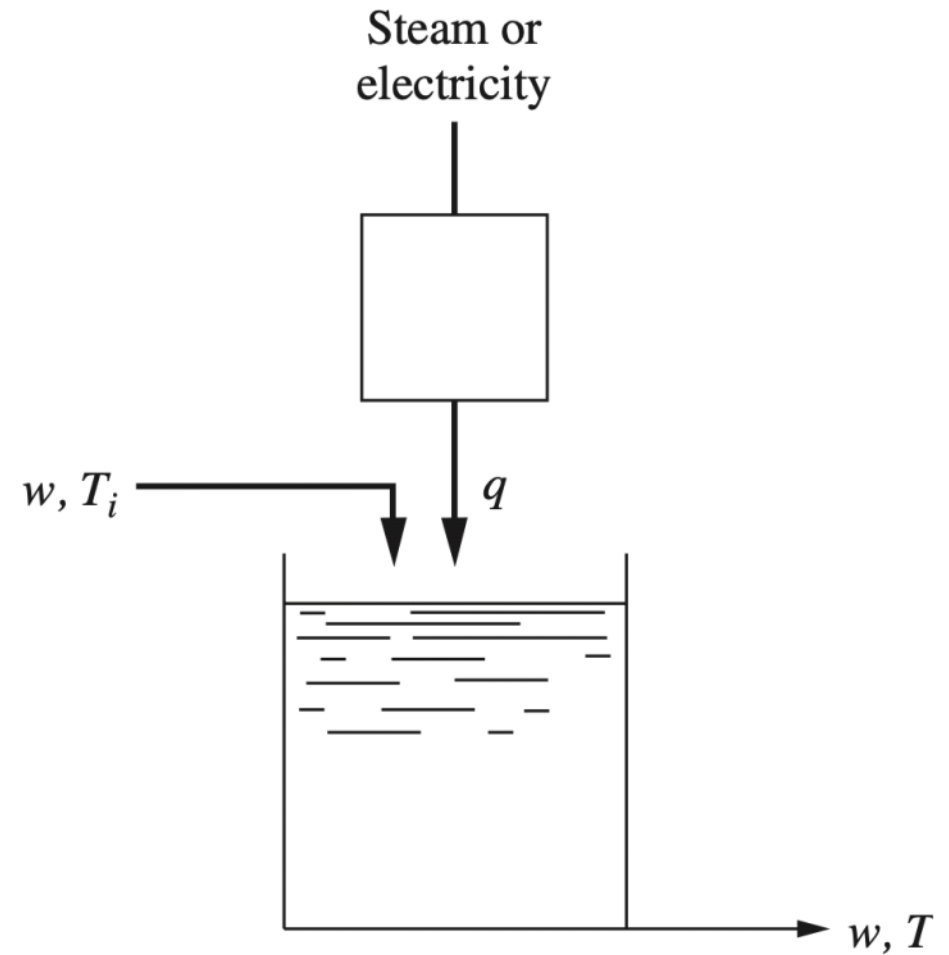
$$wC(T_i - T_{is}) - wC(T - T_s) + q - q_s = \rho VC \frac{d(T - T_s)}{dt}$$

$\downarrow$
 $T_i = T_{is} \rightarrow$ const

$\downarrow$
 T'

$\downarrow$
 Q

$\downarrow$
 T'



In deviation variable form

$$-wCT' + Q = \rho VC \frac{dT'}{dt}$$

Taking Laplace

$$-wC T'(s) + Q(s) = \rho VC (sT'(s) - T'(0))$$

$$-wC T'(s) - \rho VC sT'(s) = -Q(s)$$

$$\frac{T'(s)}{Q(s)} = \frac{1}{wC + \rho VC s}$$

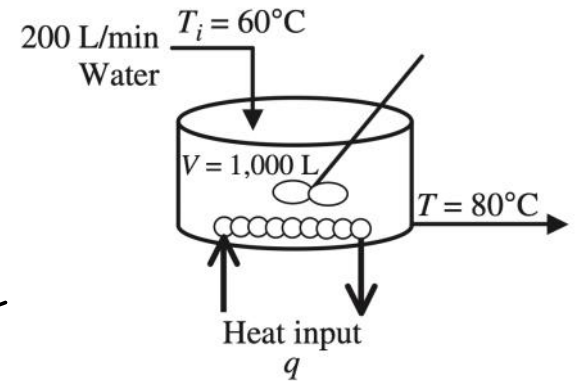
$$= \frac{1/wC}{\frac{\rho VC}{wC} s + 1}$$

$$= \frac{K}{\tau s + 1}$$

$$\frac{1}{wC} = K ; \quad \frac{\rho V}{w} = \tau$$

Example

Determine outlet temp based on inlet temp change of 10°C
 $\Rightarrow$ here $q' = q/s$



$$wC T_i' - wC T' = \rho VC \frac{dT'}{dt}$$

$$wC T_i'(s) - wC T'(s) = \rho VC (sT'(s))$$

$$\frac{T'(s)}{T_i'(s)} = \frac{wC}{\rho VC s + wC} = \frac{1}{\frac{\rho V}{w} s + 1}$$

$$\frac{T'(s)}{T_i'(s)} = \frac{1}{\tau s + 1} \quad \tau = \frac{\rho V}{w} = \frac{[\text{Kg}] [\text{m}^3]}{[\text{m}^3] [\text{Kg}]} \frac{[\text{m}^3]}{[\text{sec}]}$$

$$\tau = \frac{\rho V}{w} = \frac{V}{w/\rho} = \frac{V}{v} = \frac{1000 \text{ L}}{200 \text{ L/min}} = 5 \text{ min}$$

$$\frac{T'(s)}{T_i'(s)} = \frac{1}{5s+1}$$

Temp increase $60 \rightarrow 70^\circ\text{C}$

$$T_i' = 10^\circ\text{C}$$

$$T_i'(s) = \frac{10}{s}$$

$$T'(s) = \frac{10}{s} \frac{1}{5s+1}$$

Invert to time domain

$$\frac{10}{s} \frac{1}{5s+1} = \frac{A}{s} + \frac{B}{5s+1}$$

$$10 = A(5s+1) + Bs$$

$$10 = s(5A+B) + A$$

$$5A+B=0 \quad ; \quad A=10$$

$$B=-50$$

$$\Rightarrow \frac{10}{s} - \frac{50}{5s+1} = \frac{10}{s} - \frac{50/s}{s+1/5}$$

$$= 10 - 10 e^{-t/5}$$

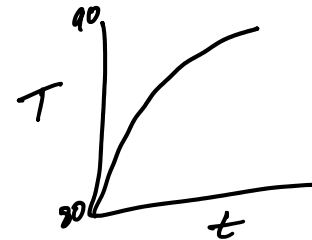
$$T'(t) = 10(1 - e^{-t/5})$$

$$\parallel$$

$$T - T_s$$

$$T(t) = T_s + 10(1 - e^{-t/5})$$

$$T(t) = 80 + 10(1 - e^{-t/5})$$



Linearization

Examples discussed thus far have been linear (linear diff eqns)
which allow transforms using Laplace

Non linear systems $\rightarrow$ approximated using linear systems

$$f(x) = f(x_s) + \left. \frac{df}{dx} \right|_{x_s} (x - x_s)$$

For liquid level system
assume flow out as
 $q_o = C h^{1/2}$
 $\uparrow$ const

Const C and Area

Material Balance

$$q(t) - q_o(t) = A \frac{dh}{dt}$$

$$q(t) - C h^{1/2} = A \frac{dh}{dt}$$

$\hookrightarrow$ Non linear diff eqn

For non linear term, $h^{1/2}$; there is no simple transform

We can linearize the non linear term

Using Taylor Series Expansion

Expanding around S.S

$$q_o = q_o(h_s) + q_o'(h_s)(h - h_s) + q_o''(h_s) \frac{(h - h_s)^2}{2} + \dots$$

$$q_o \approx q_o(h_s) + q_o'(h_s)(h - h_s)$$

q'_0 @ h_s

$$q'_0(h_s) = \frac{1}{2} C h_s^{-1/2}$$

Plugging back

$$q_0 = \underbrace{q_0(h_s)}_{q_{0s}} + \underbrace{\frac{1}{2} C h_s^{-1/2}}_{1/R_1} (h - h_s)$$

$$q_0 = q_{0s} + \frac{1}{R_1} (h - h_s)$$

Plugging into initial eqn

$$q(t) - q_0(t) = A \frac{dh}{dt}$$

$$q - q_{0s} - \frac{h - h_s}{R_1} = A \frac{dh}{dt}$$

q_s @ S.S. IN=OUT

$$q - q_s - \frac{h - h_s}{R_1} = A \frac{dh}{dt}$$

$$q - q_s = A \frac{dh}{dt} + \frac{h - h_s}{R_1}$$

$$Q = A \frac{dH}{dt} + \frac{H}{R_1}$$

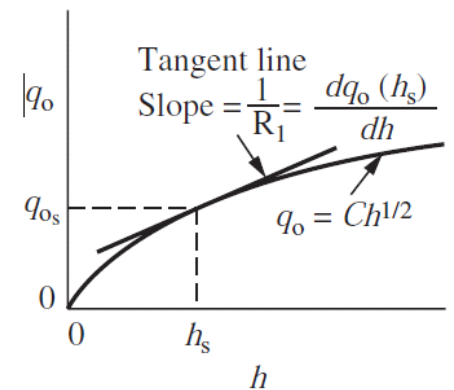
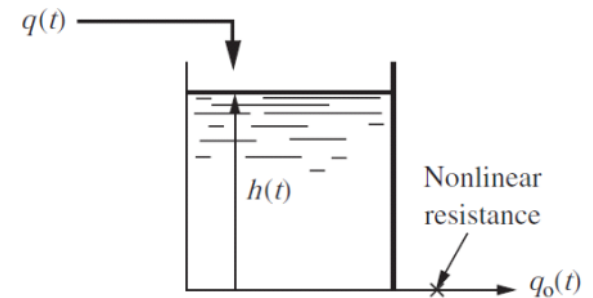
Laplace

$$Q(s) = A s H(s) + \frac{1}{R_1} H(s)$$

$$Q(s) = H(s) \left(A s + \frac{1}{R_1} \right)$$

$$\frac{H(s)}{Q(s)} = \frac{1}{A s + \frac{1}{R_1}} = \frac{R_1}{R_1 A s + 1} = \frac{R_1}{\tau s + 1}$$

$$\tau = R_1 A = \left(\frac{2 h_s^{1/2}}{C} \right) A$$



Linearization using Taylor Expansion

Diff eqn for system

$$\frac{dy}{dt} + \underbrace{f(y)}_{\text{non linear term}} = x(t)$$

$$f(x) = f(x_s) + \left. \frac{df}{dx} \right|_{x_s} (x - x_s) + \text{higher order terms}$$

$$\frac{dy}{dt} + f(y_s) + \left. \frac{\partial f}{\partial y} \right|_{y_s} (y - y_s) = x(t)$$

$$\Rightarrow \text{s.s} \Rightarrow \frac{dy_s}{dt} + f(y_s) + \left. \frac{\partial f}{\partial y} \right|_{y_s} (y_s - y_s) = x_s$$

Subtracting

$$\frac{d(y - y_s)}{dt} + \left. \frac{\partial f}{\partial y} \right|_{y_s} (y - y_s) = x - x_s$$

$$\frac{dY}{dt} + \left. \frac{\partial f}{\partial y} \right|_{y_s} Y = X$$

Problem

Derive $\frac{H(s)}{Q(s)}$ with $h_s = 1 \text{ ft}$

$$\frac{H(s)}{Q(s)} = \frac{1}{A_s} = \frac{1}{s} \quad q(t) - q_0 = \frac{A dh}{dt}$$

@ $h_s = 3 \text{ ft}$

$$q(t) - q_0 = A \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = q(t) - (6 + 2h)$$

$$q_0 = q_{0 \text{ pump}} + q_{0 \text{ valve}}$$

$$= 10 \text{ cfm} + \frac{h-2}{R}$$

$$= 10 + \frac{h-2}{\frac{1}{2}} = 10 + 2(h-2)$$

$$= 10 + 2h - 4 = 6 + 2h$$

$$\Rightarrow \text{s.s} \Rightarrow 0 = q_s - 12$$

$$q_s = 12 \text{ cfm}$$

$$\frac{dH}{dt} = Q - 2H$$

$$sH(s) + H(0) = Q(s) - 2H(s)$$

$$sH(s) + 2H(s) = Q(s)$$

$$H(s)(s+2) = Q(s)$$

$$\frac{H(s)}{Q(s)} = \frac{1}{s+2}$$

$$@ h_s = 3 \text{ ft}^3$$

$$= \frac{1}{s}$$

$$@ h_s = 1 \text{ ft}^3$$

Name: _____

5.2. A liquid-level system, such as the one shown in Fig. 5–1, has a cross-sectional area of 3.0 ft^2 . The valve characteristics are

$$q = 8\sqrt{h}$$

where q = flow rate, cfm, and h = level above the valve, ft. Calculate the time constant for this system if the average operating level above the valve is

(a) 3 ft

(b) 9 ft

Tips:

For pasting pictures, try taking screenshots using "Windows+Shift+S" and pasting directly on here

For adding word or pdf documents, add them here and right click to select "Insert as Printout"

ANSWER HERE